
DAMSIC

An Open-Source Summary

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1 Sample & Hold

This summary isn't a perfect 100% of the class, it is giving the broad idea of the class without always going over all the details and implementation.

1.1 Introduction

Doing analog to digital conversion requires 2 type of discretization :

1. Time : sampling
2. Amplitude : quantization

adc to dac conversion

r0.5

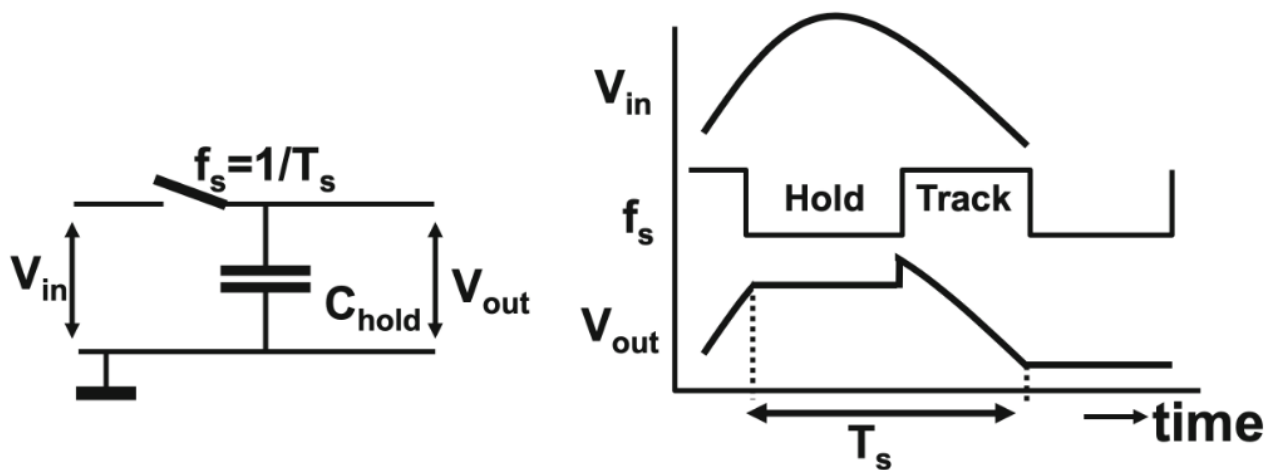


Figure 1: image

And of course, going from digital to analog requires to do some amplitude restoration and holding the signal to make it continuous. Important to understand that the digital values only make sense if we have a sampling frequency and a reference voltage where $111...111 = V_{ref}$. So the binary number represents a fraction of this reference voltage.

To sample this we are using a *Track and Hold* rather than a sample and hold since it is not quite easy to sample (opening the switch for a small small amount of time). Of course, this switch is implemented as a MOS transistor where we can take advantage of its symmetry to either charge or discharge the capacitance. If we want to have a *Sample and Hold* we can chain two track and hold back to back using inverted clocks. But it can be quite tricky to have such precision when designing the circuit.

1.1.0.1 Requirements We want good *speed*, so short tracking period and have a high bandwidth which implies to have a small τ (big switch, small cap). On the other hand, for accuracy we want to have a low noise ($\frac{kT}{C}$) so a big capacitance and low distortion so a linear switch. We can see this is a major trade-off in T&H design.

1.2 Non-idealities

1.2.1 Noise

r0.33

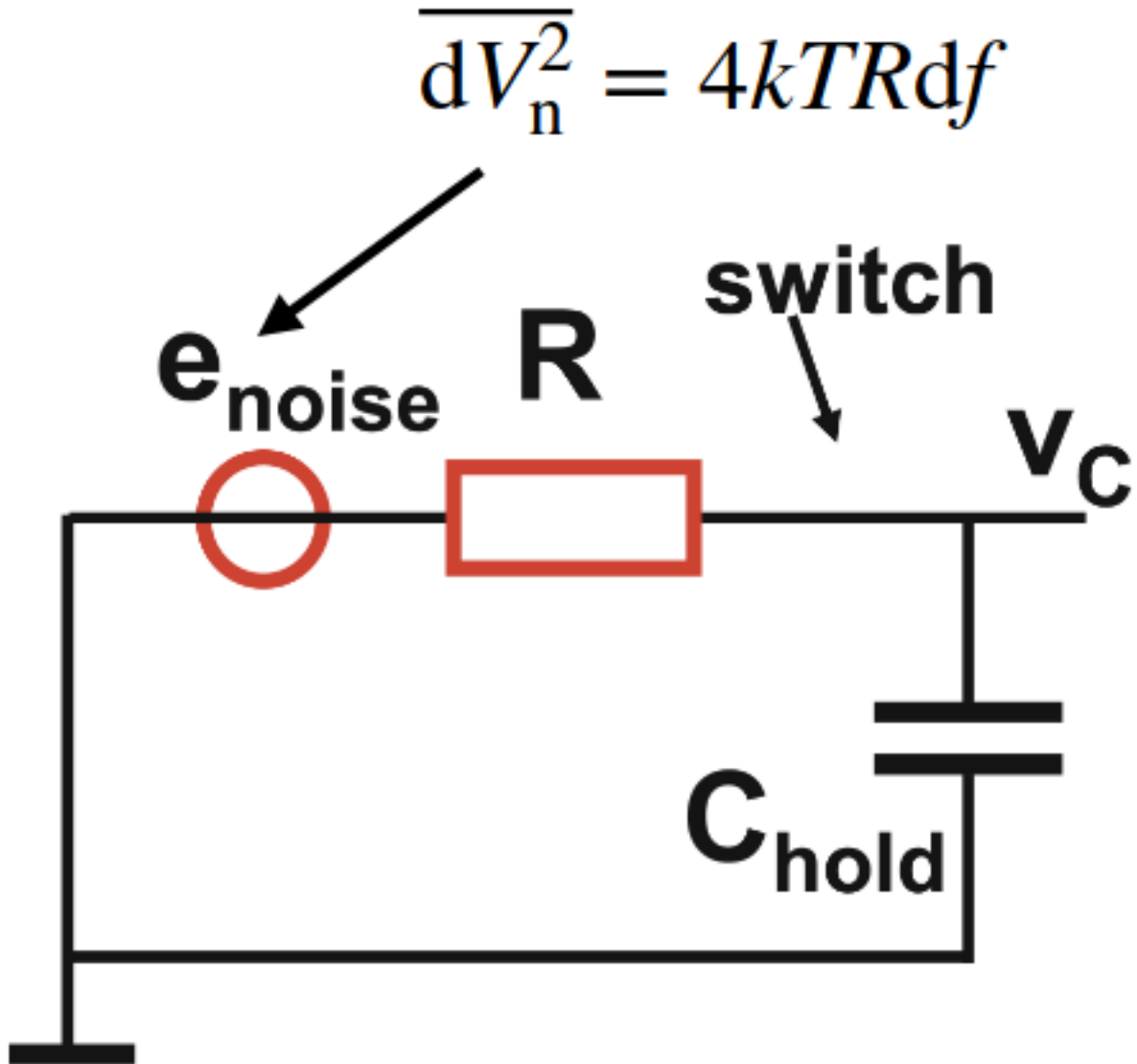


Figure 2: image

We can represent the switch (implement as a transistor) with a resistor that has an equivalent noise source e_{noise} . Depending on the C_{hold} value we will have various difference equivalent noise voltages:

$$v_{C,noise}^2 = \int_{f=0}^{f=\infty} \frac{4kTRdf}{1 + (2\pi f)^2 R^2 C^2} = \frac{kT}{C}$$

$$SNR = \frac{P_{signal}}{P_{noise}} = \frac{A^2/2}{kT/C}$$

Where $A/\sqrt{2}$ represents the RMS amplitude of the signal. It is interesting to notice how this noise is only present when we are having the switch high so theoretically no noise should be present in the hold phase.

Hence, this noise is also sampled ! Even though it will be flat for any frequency below Shannon and has a zero mean and variance of $\sigma = kT/C$. This will create some *aliasing*. The noise PSD will **increase** due to sampling since we will have some shorter bandwidth (*check this*).

$$S_{vv,RC} = 4kTR = \frac{2kT}{\pi C f_{RC}} \quad S_{vv,s} = \frac{2kT}{C f_s}$$

What happens is the fact we have a noise density covering all possible frequencies which gets sampled and so we have a stack up of noise which is technically infinite. But since we have a capacitance and resistor that samples the signal and noise this will act as a low pass filter and will create some sorts of lobes. The **total sampled noise** is always kT/C .

The increase of noise spectral density depends on noise bandwidth and this causes problem for high-resolution converters. It is not always ideal to have a high bandwidth

1.2.2 Jitter

It is a form of *statistical noise* and is due to noisy clock generator which won't sample reliably and will sometimes happen slightly before or after the "tick". Jitter can be deterministic which will result in *tones* in the spectrum or distortion for signal-dependent jitter (definition of what distortion is). We can find the result of σ_t has on our sampled amplitude :

$$\begin{aligned} A(t) &= A \sin(\omega t) & \sigma_A^2(nT_s) &= \left(\frac{dA(nT_s)}{dt} \right)^2 \sigma_t^2 = \omega^2 A^2 \cos^2(\omega nT_s) \sigma_t^2 \\ A(nT_s + \Delta t(t)) &= A \sin(\omega(nT_s + \Delta t(t))) & \sigma_A^2 &= \frac{\omega^2 A^2 \sigma_t^2}{2} \\ \Delta A(nT_s) &= \frac{dA \sin(\omega t)}{dt} \Delta t(nT_s) & \Delta A(nT_s) &= \omega A \cos(\omega nT_s) \Delta t(nT_s) \end{aligned}$$

The jitter acts as a sort of ceiling, to fight against the thermal noise we can simply increase the input voltage but at some point the SNR due to noise and jitter are equal and the SNR of jitter is the limit since it only depends on the frequency.

1.2.3 Non fundamental error - Implementation error

1.2.3.1 Pedestal Pedestal error is linked with the charges in the channel **and** the parasitic cap C_{ped} which goes from gate to source. There is some charges that once the switch is closed will get injected or removed. One solution is the compensation circuit where we will use dummy switches which will take those extra charges. This requires a duty cycle of 50% and precise clock to "capture" those extra charges precisely.

1.2.3.2 Droop The charges put on C_{hold} may go down with time especially with BJT as the base will draw currents. It is linked with leakage :

$$V_{droop} = -\frac{I_{leak} T_{hold}}{C_{hold}}$$

1.2.3.3 Feedthrough We also have a capacitance between the source and the drain which will cause a small current to *feedthrough*. One solution is to use **t-switch** which helps to reduce the effect. In differential implementations we can use some *cross-coupled always-off* switches.

1.2.3.4 Aperture Jitter We won't have a perfect instantaneous rise so the logic 1 or logic 0 may not appear exactly at the clock period.

1.3 Capacitor and switch implementations

1.3.1 Capacitors

In IC design we will mostly focus on **Metal-Insulator-Metal** (MIM) capacitor which requires a special process and are linear. The other type is the interconnect **MOM** which doesn't require a special process. One key idea is to create the capacitance as high as possible on the metal layers to get away from parasitic due to the wafer.

We can also do some *vertical* MOM. This solution takes advantage of the height of the metal stack but the lower end will be more susceptible to parasitic and we use the other symbol of a capacitance.

It is sometimes hard to model capacitance due to the fact we have multiple field lines going and connecting with all the layers. So one trick is to isolate each capacitance and shield each one of them. It makes analysis much easier.

1.3.2 Switches

The switch (NMOS or PMOS) has its own R_{on} which is proportional to $1/V_{in}$. We not only need a small enough resistor for good τ but also a constant one to avoid distortion !

$$R_{on,NMOS} = \frac{1}{(W/L)_N \beta_{N\Box} (V_{DD} - V_{in} - V_{T,N})} \quad R_{on,PMOS} = \frac{1}{(W/L)_P \beta_{P\Box} (V_{in} - |V_{T,P}|)}$$

One idea to reduce this $\propto V_{in}$ is to use **complementary switches** so the NMOS and PMOS behavior will kinda cancel out. For this to work we need sufficient V_{DD} namely $V_{DD} < V_{th,n} + |V_{th,p}|$. We can boost V using some *Cockroft-Walton multiplier* or *Dickson multiplier*.

1.3.2.1 Bootstrapping The idea behind bootstrapping is to keep the sample pulse and input voltage constant which will result in constant R_{on} .

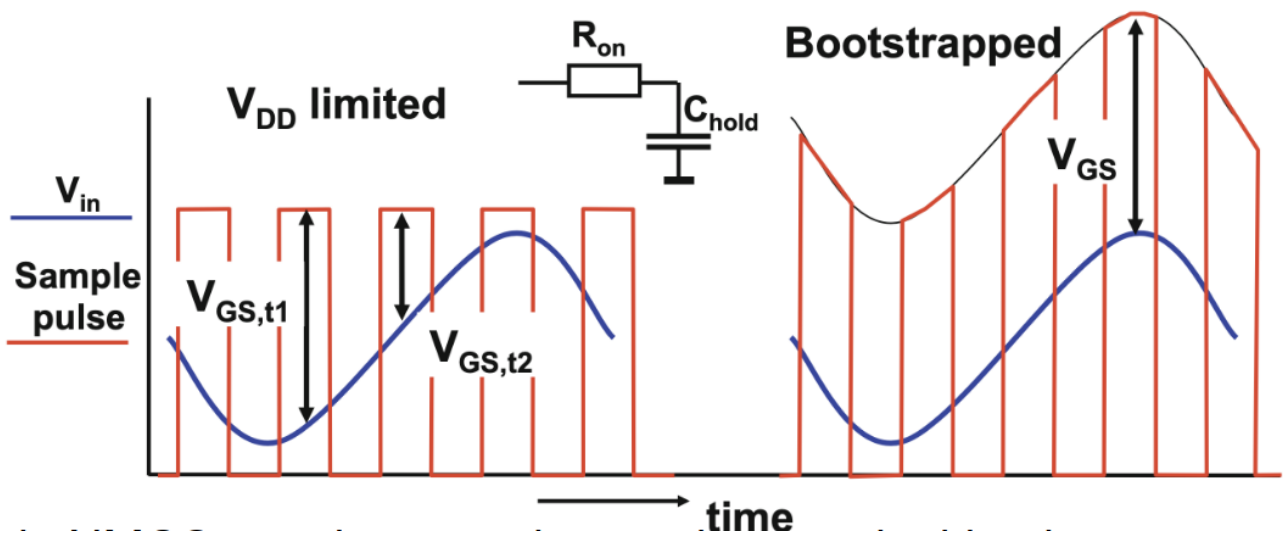


Figure 3: Bootstrapping

Here we don't need to use two transistor, we can just use one NMOS and we get a high SNR for small C_{hold} . We keep the resistance constant and we don't have any distortion.

The main idea is to use a capacitance we charge to V_{DD} in hold phase and when the tracking phase come the V_{in} is added on top of the V_{DD} that is stored on the cap making the $V_{gs} = V_{DD}$.

Some challenges we can face are voltage limit across the gate, drain and source. We also have a body diode which can create charge lost, latch-up or even current to substrate.

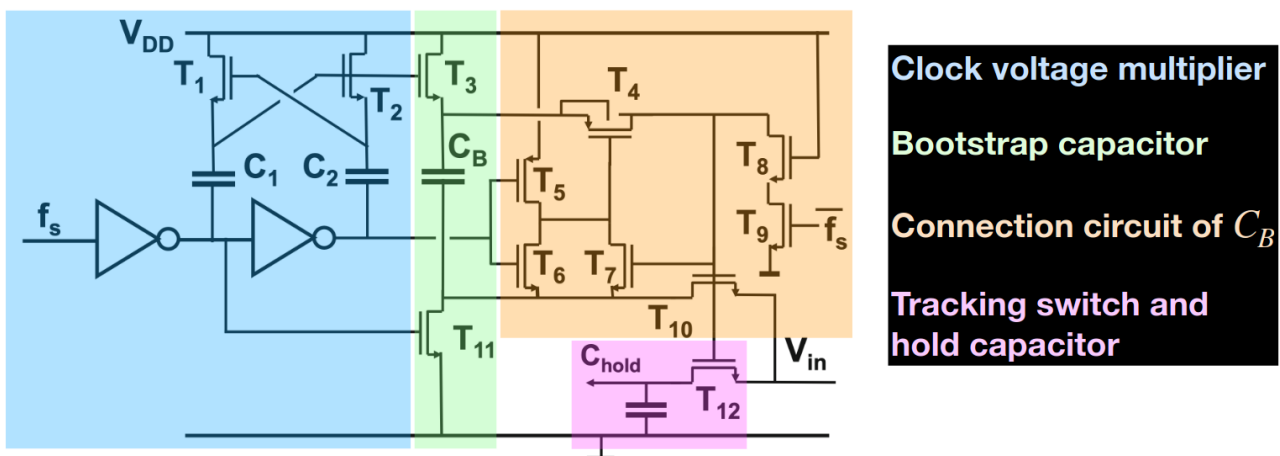


Figure 4: Typical bootstrap circuit

We can also use a technique called *bottom-plate sampling*. add more info read a bit more about it.

1.4 Circuit topologies

1.4.1 General considerations

We want to maintain the value on the capacitance, one solution is to use buffered implementation. Buffered implementation will be more accurate but we introduce a feedback system which inherently has delay thus some maximum possible speed.

We also need to keep in mind that when using a buffer (opamp) we will have an extra *non-linear* capacitance in parallel with our hold capacitance. So we need to pick the right hold capacitance to minimize this extra non linear capacitance.

We always need to keep in mind what is this designed use for, what is the swing (rail-to-rail, buffer supply voltage above V_{DD} , ...).

High-gain opamp requires a low settling error $A_{DC} > \frac{1}{\epsilon} = 2^N$. The settling speed determines unity-gain frequency of a track and hold buffer. the UGF plays a big role.

1.4.2 Switched-capacitor T&H circuits

The most basic switched-cap implementation :

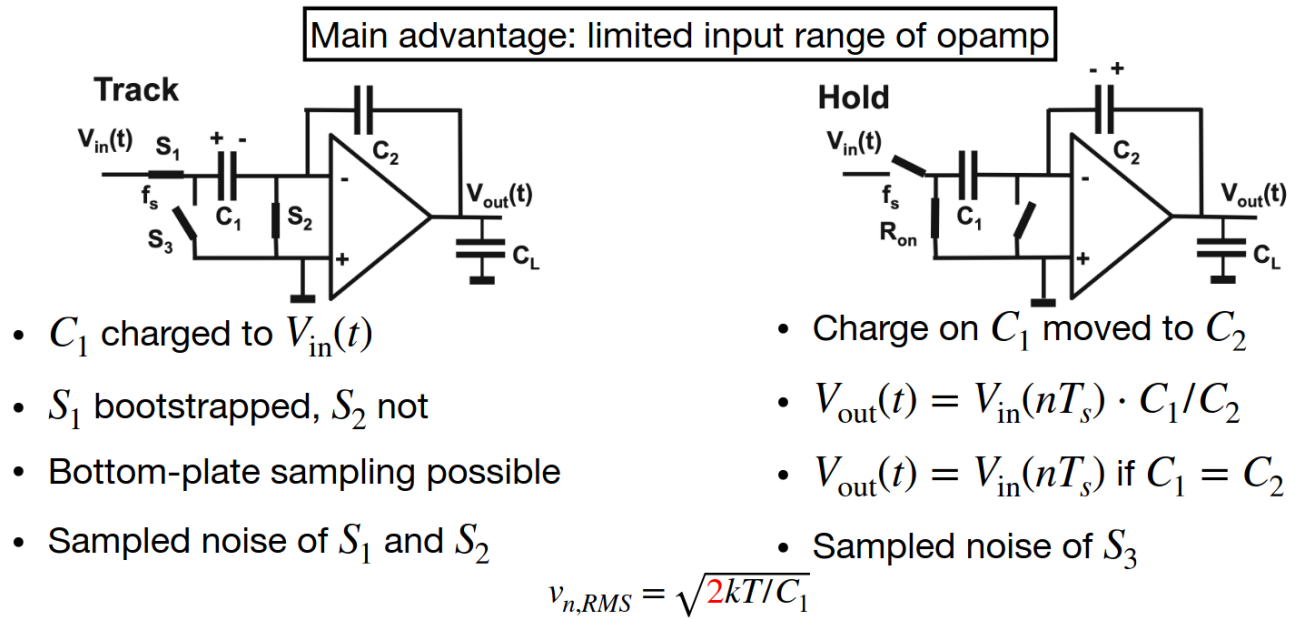
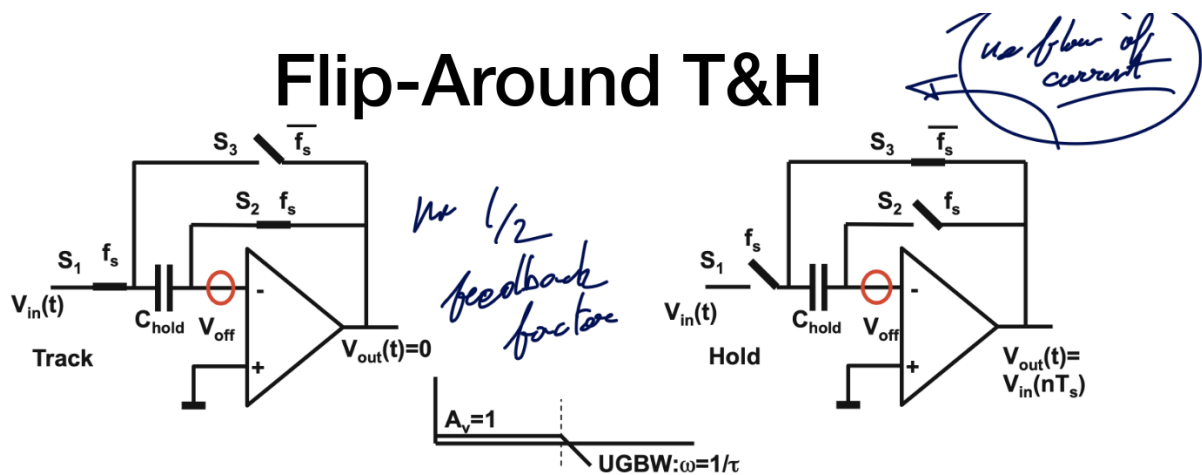


Figure 5: Basic Switched-cap implementation

In the *track phase* we first discharge C_2 . The S_1 is *bootstrapped* and C_1 is charged to V_{in} . We have some sampled noise at S_1 and S_2 . In the *hold phase*, the charge from C_1 are moved to C_2 (charge recollection). We have $V_{out}(t) = V_{in}(nT_s) \cdot C_1/C_2$.

If we have *unity-gain signal* we will have a $H = C_2/(C_1 + C_2) = 1/2$ if $C_1 = C_2$. The closed loop gain is $A_{cl} = 2$ so we will amplify some unwanted signal. The setting time is determined by $1/2\tau$.

1.4.3 Flip-Around T&H



- Opamp in unity-gain feedback
- C_{hold} charged to $V_{in}(t)$
- S_1 bootstrapped, S_2 not
- Bottom-plate sampling possible
- Opamp in unity-gain feedback → faster than the previous topology
- No amplification of input noise
- $V_{out}(t) = V_{in}(nT_s)$

Figure 6: Flip-Around track & hold

It will use a virtual ground to charge and discharge the cap. Here we have a unity-gain feedback of 1 which is desirable. But this virtual ground isn't perfect and so we will have some *offset*. But this offset will actually be cancelled !

$$\text{Track mode : } V_{C_{hold}}(nT_s) = V_{in}(nT_s) - V_{off}$$

$$\text{Hold mode : } V_{out}((n + 1/2)T_s) = V_{C_{hold}}(nT_s) + V_{off} = V_{in}(nT_s)$$

If we go in the z-domain to analyze this error we find :

$$V_{out,error} = V_{off}(z)(1 - z^{-0.5})$$

$$H(f) = 2\sin(\pi f/2f_s)$$

So it will suppress the offset and the noise at low-frequency. But the high-frequency noise will be doubled and the sample noise on C_{hold} is still present.

Amplifying T&H Circuit

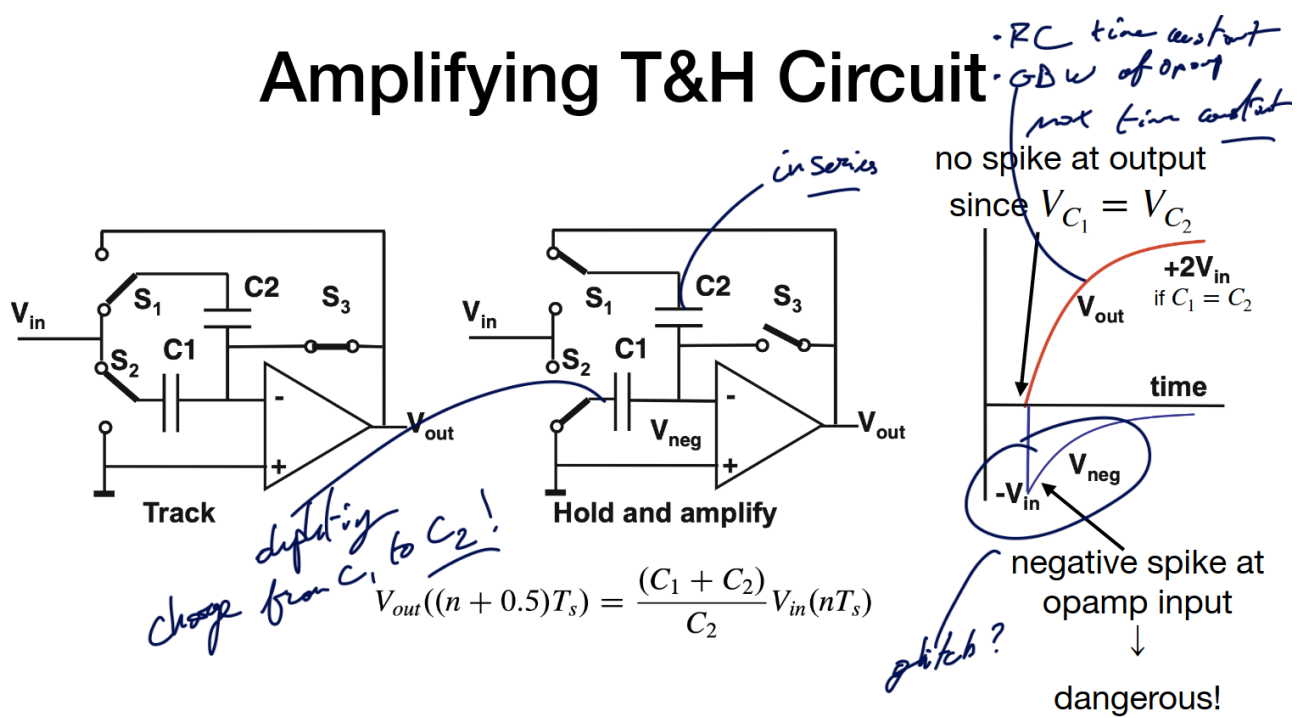


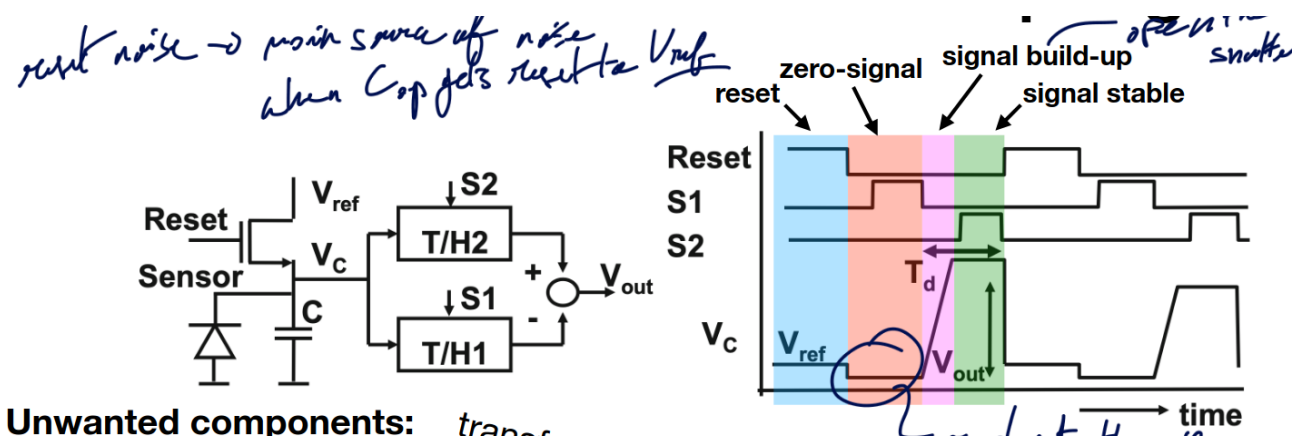
Figure 7: Amplifying track & hold

We can see some dangerous spikes on fig. 1.5. The noise can be found as follow :

$$v_{in,noise}^2 = \frac{kT}{C_1 + C_2} \quad v_{out,noise}^2 = \frac{kT(C_1 + C_2)}{C_2^2} + \text{noise OpAmp}$$

1.4.4 Correlated double sampling

In sensors and image sensor we want to get rid of those unwanted component. Often, we will have 2 samples time where we will first get a baseline then the actual signal we want to remove the offset component.



Unwanted components: trans

Figure 8: Correlated double sampling

$$|H(s)| = |1 - e^{-sT_d}| = |e^{-sT_d/2}| = |e^{-sT_d/2} 2\sin(sT_d/2)|$$

$$|H(\omega)| = |2\sin(\omega T_d/2)| = |2\sin(\pi f T_d)|$$

It will reject the low-frequency noise but will *amplify* by $1/2T_d$ the noise near *odd multiples*.

2 Digital-to-Analog Conversion

We need to differentiate two different use cases of a dac namely :

1. Convert to physical domain : when we need a *high quality* at every time instant. We need a minimum amount of power to be able to draw some current. For example a USB-C to jack adapter !
2. Generate reference value for an adc : typically needed in a SAR adc. We only need this conversion at some specific point in time and the drive capabilities are not exigent.

2.1 Signal representations

- Straight binary: only positive signals
- Two's complement: easy addition and subtraction, zero signal at half reference (noisy!)
- Sign and magnitude: no noisy half reference values for zero signal, digital decoder needed, simple rounding results in cross-over distortion
- Gray coded: only one bit changes

Table 1: Binary Representations

Straight binary		Two's complement		Sign+magnitude		Gray coded	
15	1111	7	0111	7	0111	15	1000
14	1110	6	0110	6	0110	14	1001
13	1101	5	0101	5	0101	13	1011
12	1100	4	0100	4	0100	12	1010
11	1011	3	0011	3	0011	11	1110
10	1010	2	0010	2	0010	10	1111
9	1001	1	0001	1	0001	9	1101
8	1000	0	0000	0	0000	8	1100
7	0111	-1	1111	0	1000	7	0100
6	0110	-2	1110	-1	1001	6	0101
5	0101	-3	1101	-2	1010	5	0111
4	0100	-4	1100	-3	1011	4	0110
3	0011	-5	1011	-4	1100	3	0010
2	0010	-6	1010	-5	1101	2	0011
1	0001	-7	1001	-6	1110	1	0001

Straight binary		Two's complement		Sign+magnitude		Gray coded	
0	0000	-8	1000	-7	1111	0	0000

To represent those bits in hardware we can go either for the *thermometer* or *binary* representation. In the thermometer representation each cells are of the same size and to increase numbers we need to increase the amount of cells on. It grants **monotonicity**. For the binary, they are not of equal size but has the advantage to need n for n bits while it is 2^n cells for thermometer. It is not guaranteed to be monotonic and the large switch of cells can create glitches.

We can also use *segmentation* to leverage from both type of implementation.

Table 2: Comparison between Unary and Binary Architectures

Architecture	Unary	Binary
Monotonicity	By design	Not guaranteed
Number of elements	$\propto 2^N$	$\propto N$
Area, parasitics	$\propto 2^N$	$\propto N$
inl systematic	Gradient	Small
inl random	$\propto \sigma_{element} \sqrt{2^N - 1}$	Same with dnl errors
dnl random	$\propto \sigma_{element}$	$\propto \sigma_{element} \sqrt{N, \dots, 2^N - 1}$
Switching energy	$\propto$ Signal	Can be large and non-linear
Decoder complexity	2^N -to-1	Simple
Power	Choice for R, I, or C implementation dominates power	
Noise	Similar for similar power levels	

We need to keep in mind that the dnl is the difference between one step to the other and it must be lower than $1LSB$. The inl is the integrate error and so cumulate the error. Those issues will lead to distortion and harmonics !

2.1.1 Typical inl of Unary Architectures

Here we have A_{lsb} that represents the minimum voltage we can add so the minimum cell. To represents a number and it's unused counter part :

$$y_1 = \sum_{i=0}^{m-1} A_{lsb}(i) \quad \mathbb{E}(y_1) = mA_{lsb} \quad \sigma_{y_1} = \sigma_{A_{lsb}} \sqrt{m}$$

$$y_2 = \sum_{i=m}^{2^N-1} A_{lsb}(i) \quad \mathbb{E}(y_2) = (2^N - m)A_{lsb} \quad \sigma_{y_2} = \sigma_{A_{lsb}} \sqrt{2^N - m}$$

To find the inl we can use this formula **WHY THIS FORMULA I DON4T GET IT ???**

$$\begin{aligned} INL(m) &= \frac{2^N \sum_{i=0}^{m-1} A_{lsb}(i)}{\sum_{i=0}^{2^N-1} A_{lsb}(i)} - m = \frac{2^N y_1}{y_1 + y_2} - m \\ \sigma_{INL}^2(m) &= \left(\frac{\partial INL(m)}{\partial y_1} \right)^2 \sigma_{y_1}^2 + \left(\frac{\partial INL(m)}{\partial y_2} \right)^2 \sigma_{y_2}^2 \\ &= \left(\frac{y_2}{(y_1 + y_2)^2} \right)^2 m \sigma_{A_{lsb}}^2 + \left(\frac{-y_1}{(y_1 + y_2)^2} \right)^2 (2^N - m) \sigma_{A_{lsb}}^2 \\ &= \frac{m(2^N - m)}{2^N} \frac{\sigma_{A_{lsb}}}{A_{lsb}^2} \end{aligned}$$

Binary DNL and INL With 255 Elements

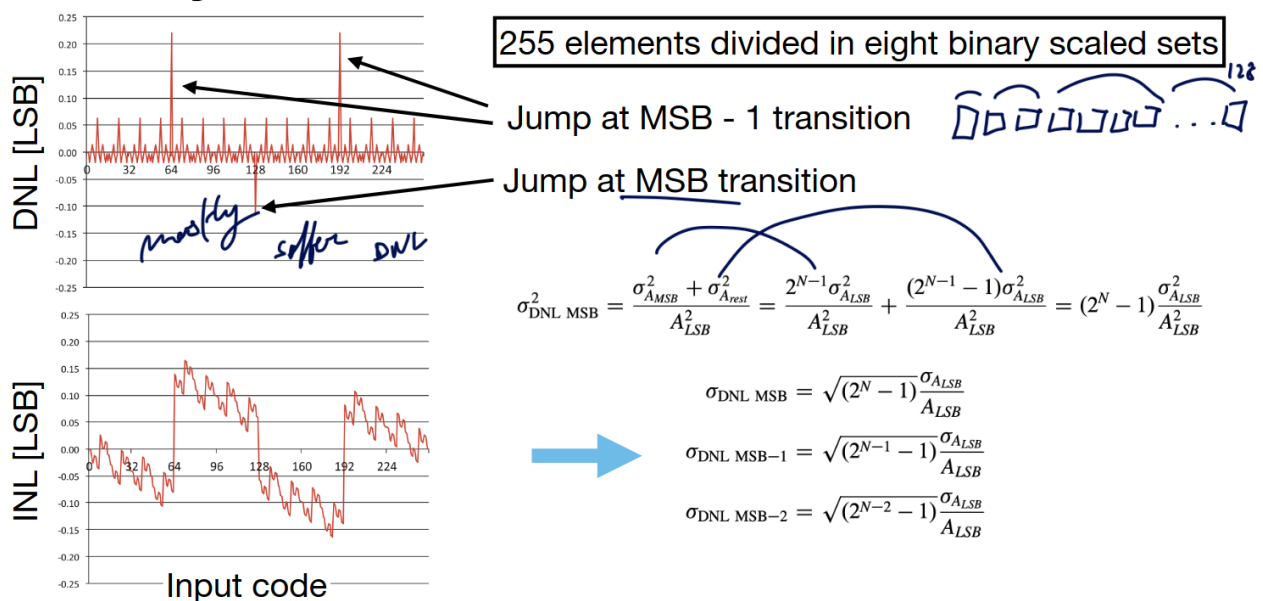


Figure 9: Segmented dnl and inl

2.1.1.1 Segmented cells The less binary scaled elements we have the stronger those jumps and glitches will be. It may even lead to dnl being over 1 LSB meaning we can't guarantee monotonicity anymore this will result in missing code.

$$\sigma_{DNL,MSB} = (2^N - 1) \frac{\sigma_{A_{lsb}}}{A_{lsb}}$$

Table 3: Comparison of Unary and Binary Architectures, dac and adc

	Unary	Binary
Voltage	Resistor string <i>Flash adc</i>	R-2R <i>Low-performance dac</i>
Current	Current matrix <i>High bandwidth dac</i>	Current splitting

	Unary	Binary
Charge/capacitor	Capacitor bank <i>Low power dac</i>	Capacitor bank
Time	PWM, $\Sigma\Delta$ mod <i>Low bandwidth dac</i>	Limited by distortion

2.2 Digital-to-analog conversion

2.2.1 Voltage domain

2.2.1.1 Resistor ladder The simplest dac we can think of is a **resistive ladder** where we use a thermometer code based approach. We need 2^N resistors between V_{ref+} and V_{ref-} . We need a buffer to drive the load so we don't have some currents that flow back into the resistive ladder which would cause some monotonicity and values change. This buffer op-amp isn't straightforward as it needs to be a rail-to-rail op-amp. The equivalent input impedance seen by the input of the buffer is given as :

$$R_{eq}(m) = \frac{\frac{m}{2^N} R_{tot} \cdot \frac{2^N - m}{2^N} R_{tot}}{\frac{m}{2^N} R_{tot} + \frac{2^N - m}{2^N} R_{tot}} = \frac{m(2^N - m)}{2^{N+1}} R_{tot}$$

We have a signal-dependent current delivery and the resistor is parabolic. Due to the capacitive load we also have a signal-dependent time constant. This lead to distortion at high frequencies.

$$\begin{aligned}
 V(m) &= \frac{m}{2^N} V_{ref} = \frac{mR}{mR + (2^N - m)R} V_{ref} = \frac{R_1}{R_1 + R_2} V_{ref} \\
 \sigma_V^2(m) &= \left(\frac{\partial V(m)}{\partial R_1} \right)^2 \sigma_{R_1}^2 + \left(\frac{\partial V(m)}{\partial R_2} \right)^2 \sigma_{R_2}^2 \\
 &= \left(\frac{R_2}{(R_1 + R_2)^2} \right)^2 \sigma_{R_1}^2 V_{ref}^2 + \left(\frac{-R_1}{(R_1 + R_2)^2} \right)^2 \sigma_{R_2}^2 V_{ref}^2 \\
 &= \frac{m(2^N - m)}{2^N} \frac{\sigma_R^2}{R^2} \frac{V_{ref}^2}{2^{2N}} = \frac{m(2^N - m)}{2^N} \frac{\sigma_R^2}{R^2} V_{LSB}^2 \\
 \sigma_{INL}^2 &= \frac{m(2^N - m)}{2^N} \frac{\sigma_R^2}{R^2}
 \end{aligned}$$

ADD MORE INFO

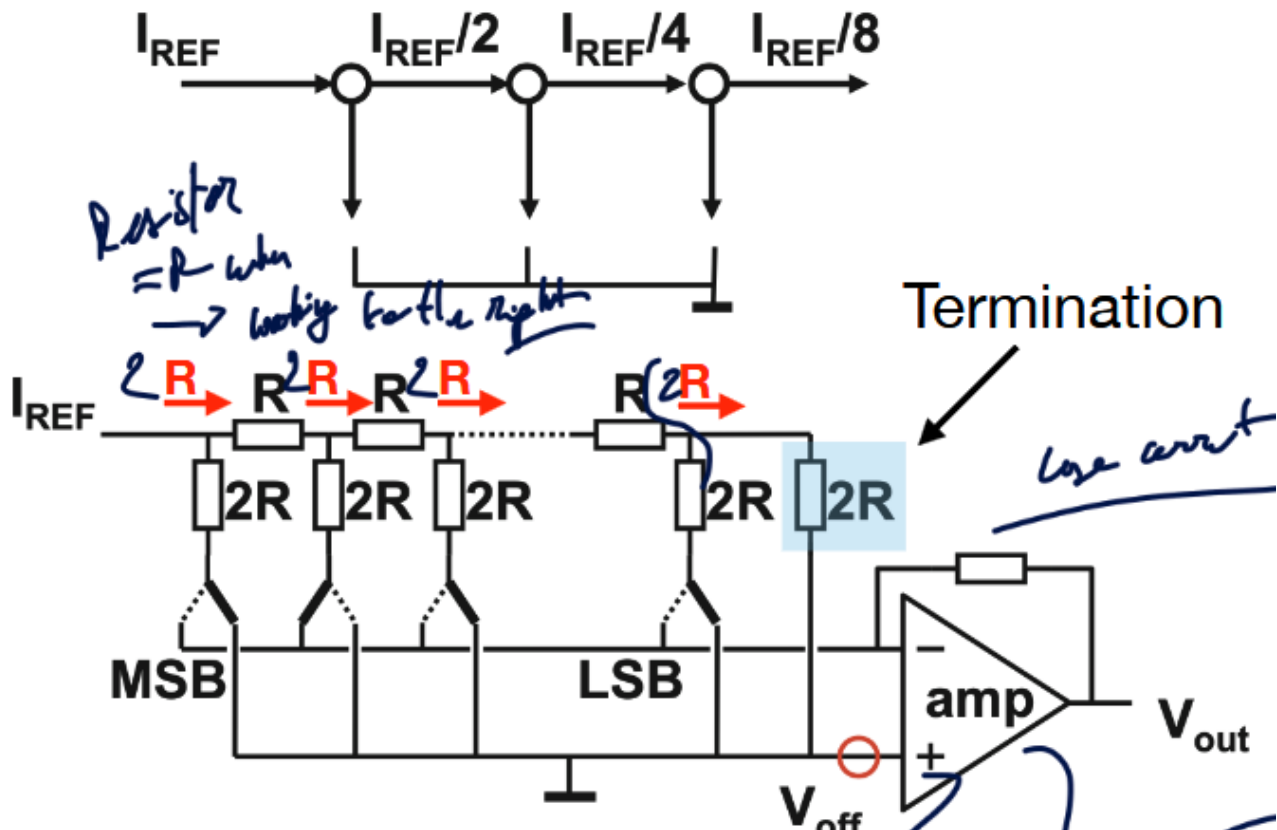


Figure 10: R-2R Ladders

2.2.1.2 R-2R ladder Here we are using a binary structure and we only need a $2N$ resistors. It is not super precise since we have a lower and lower current flowing in each branch which will be more susceptible to noise. The accuracy depends on :

1. Offset voltage : non-ideal virtual ground and so we have some error in the current split. $V_{off} < \frac{I_{REF}R}{2^N}$.
2. Resistor matching : we need good matching or we will have large difference.
3. Switch resistance : the switch will add some R_{on} resistance so we need it to be small to have a small impact.

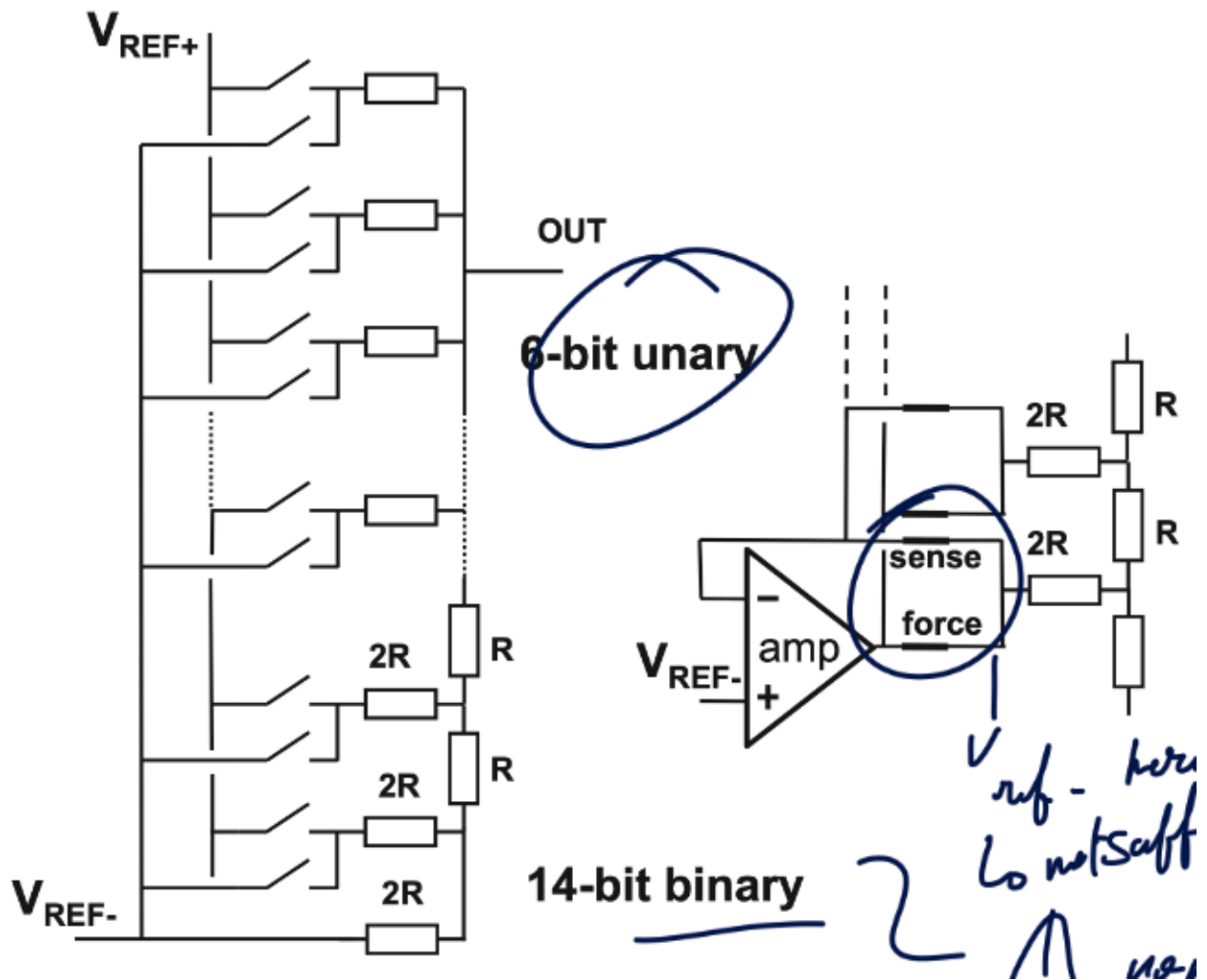


Figure 11: Segmented resistor ladder

2.2.1.3 Segmented resistor ladder We will have a thermometer and binary part. It will improve the precision. There is no current in sense path. The sense node is equal to V_{ref} . We calibrate this offset voltage and we can also call it a *kelvin contact* or *four-point sensing*.

2.2.2 Current domain

Here instead of using resistor that will drive another resistor creating a voltage that gets buffered, we will directly work with current source that will drive a resistor which will have its voltage buffered. We can either go for a thermometer or binary coded version :

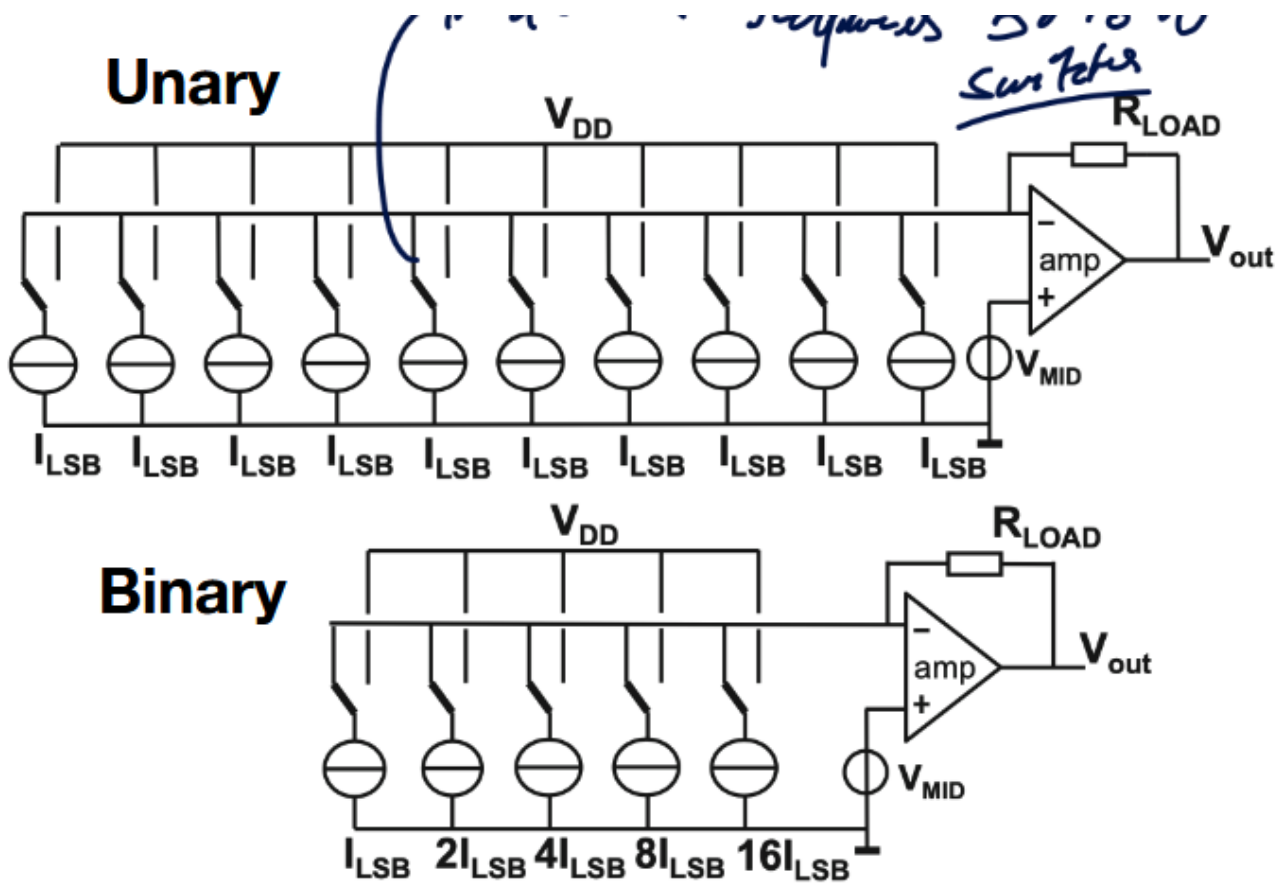


Figure 12: Buffered Current dac

We need to keep all current source on even if they are not used to avoid *current dump*. The amplifier is sensing the current so we call it a *tia*. We need a low input impedance to limit the swing and low output one so we can drive any type of load.

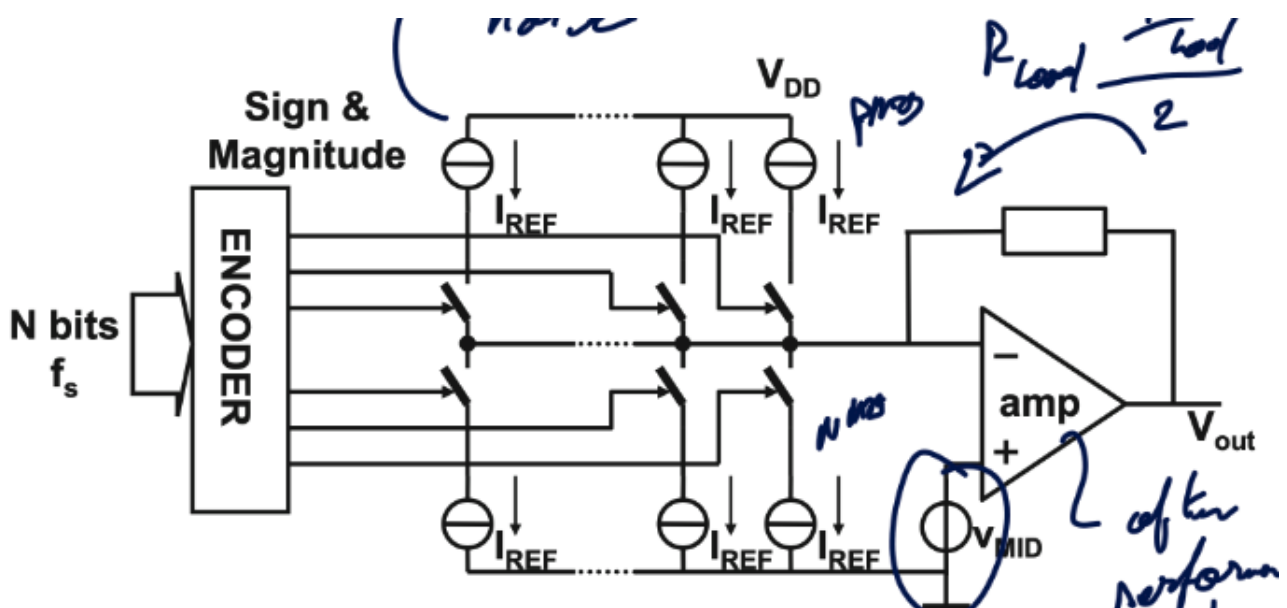


Figure 13: Symmetrical buffered current-DAC

2.2.2.1 Symmetrical Buffered current-DAC To avoid some important switching (on and off) due to the representation of a number in signed format, we want to use a symmetrical one. Since most of the signal will be around 0 often going positive and negative. This configuration will reduce the $1/f$ and thermal noise. Here we need a current sink and current source so a NMOS and PMOS network is required (may be a designed constraint and we need to match PMOS and NMOS). We only have the noise of the buffer for zero level. We will need some calibration to manage a balance between NMOS and PMOS.

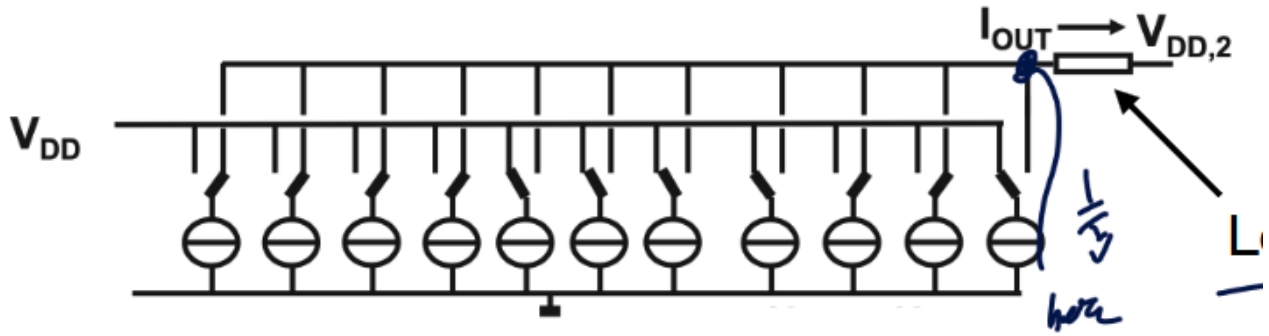


Figure 14: Buffer-less Current DAC

2.2.2.2 Buffer-less current-DAC Since we are not using a buffer, we have theoretically a larger bandwidth since we are not limited by the spec of a t_{ia}. But in life there is always parasitic and so we will have a pole determined by the output load impedance and capacitance. It is a good choice for time-continuous signals with a matched impedance between 50 – 75Ω.

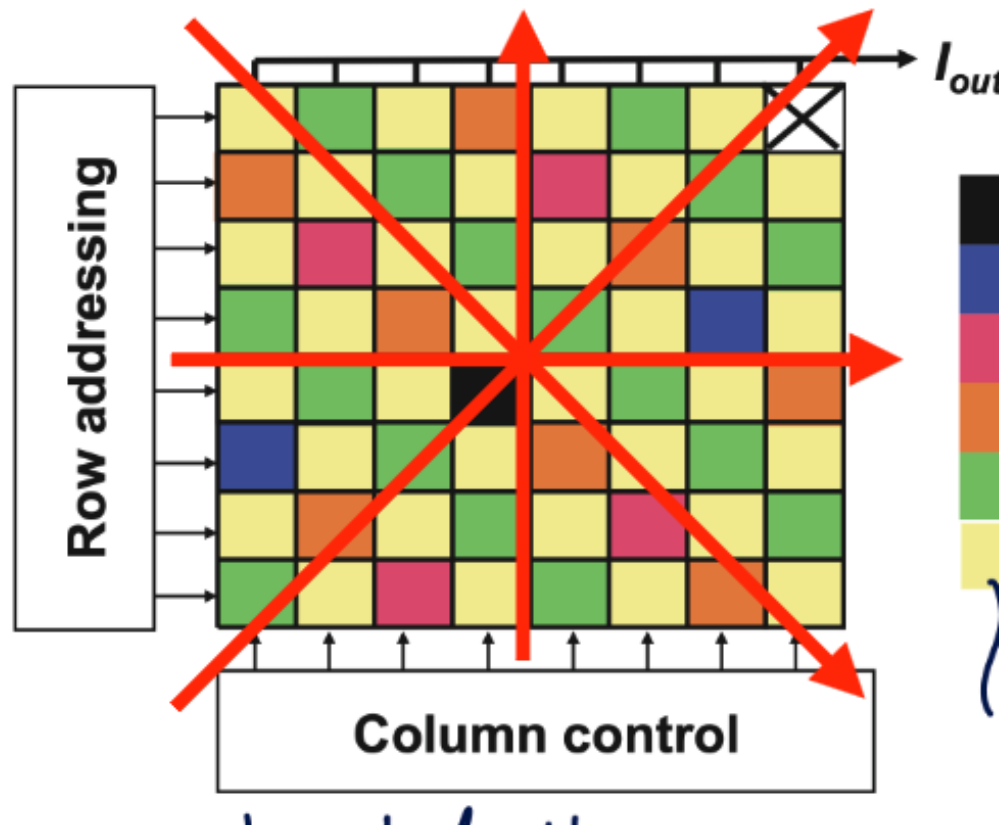
2.2.2.3 Segment current DAC Based on the same idea of a buffer-less DAC we can use some current source that are 2^a LSB and some binary based sources starting at 1. This is segmenting and we use the thermometer for MSB part and the binary for the LSB as they take less space. The biggest issue is when we switch off binary section and we switch to the next unary source :

$$\sigma_{\text{one DNL}}^2 = \frac{\sigma_{I_{\text{unary}}}^2 + \sigma_{I_{\text{all binary}}}^2}{I_{\text{LSB}}^2} = 2^{2N_{\text{binary}}} \frac{\sigma_{I_{\text{unary}}}^2}{I_{\text{unary}}^2} + (2^{N_{\text{binary}}} - 1)^2 \frac{\sigma_{I_{\text{all binary}}}^2}{I_{\text{all binary}}^2} = \frac{2A_{VT}^2 2^{2N_{\text{binary}}}}{(WL)_{\text{unary}} (V_{GS} - V_T)^2} + \frac{2A_{VT}^2 (2^{N_{\text{binary}}} - 1)^2}{(WL)_{\text{all binary}} (V_{GS} - V_T)^2}$$

Area of binary section equal to area of one unary source $\rightarrow \approx 2^{N_{\text{binary}} + 0.5} \frac{\sigma_{I_{\text{unary}}}}{I_{\text{unary}}} \approx \frac{2A_{VT} 2^{N_{\text{binary}}}}{\sqrt{(WL)_{\text{unary}} (V_{GS} - V_T)}}$

Figure 15: image

We can also go further and add an extra rail to create current divider. This third rail will *steer* unary current to current divider. The divider can be unary or binary scaled but we will have large transition error at the edge that will be reduced.

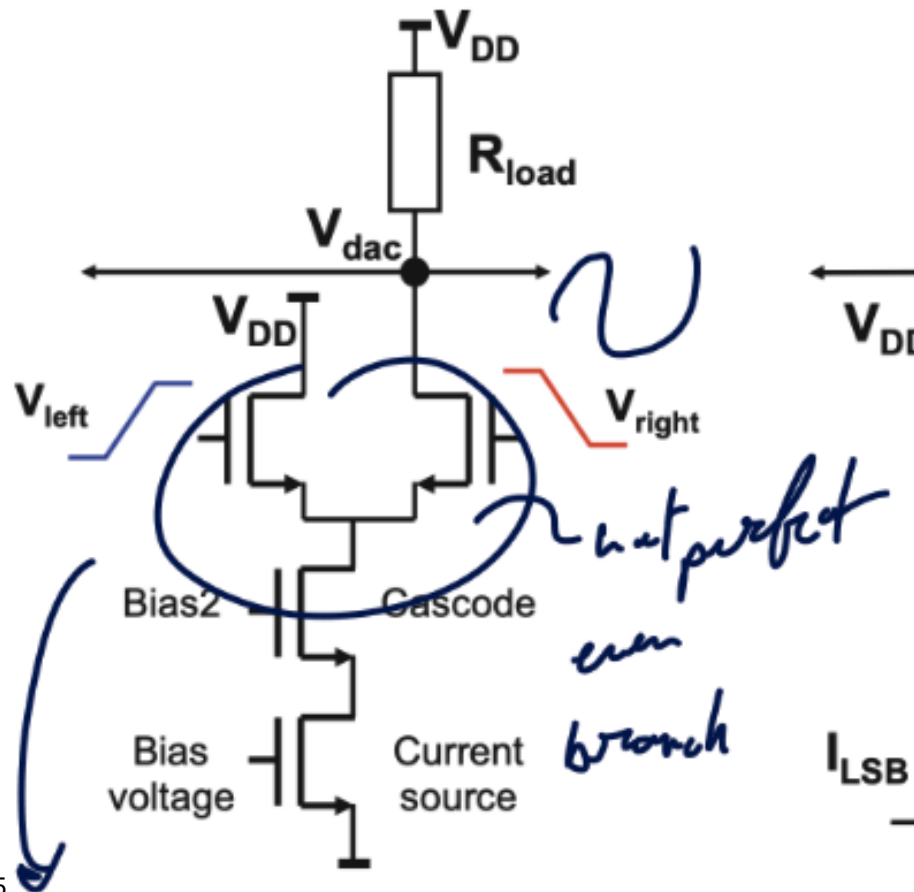


2.2.2.4 Matrix decoding r0.5

Usually in VLSI we will arrange thermometer current source in a matrix format so we can simply and quickly access the current. But due to the spreading of those current sources, we will accentuate some physical phenomena such as *oxide thickness*, *power delivery drop*, *clock timing*, *temperature*, ... which will make the current sources unequal.

One technique to reduce this effect is presented on figure [fig:centroid-layout-label]. We create symmetry in both direction and we will get rid of any **linear gradient**. We can go even further with some Q^2 walk and distributed subcell across the complete array or even use some *randomize current cell* from one sample to the next (more advanced).

Some calibration we can make is **current source sorting** where we will measure all of the cells and create a sort of pair where we trigger a larger cell with a smaller one after. But we need a lot of pre-processing and measurements and we need to create some special metal routing over the cells which can deteriorate the performances even more. We also need into account possible delay and timing issues, ...



2.2.2.5 Current cell implementation r0.5

We keep the current source on to avoid distortion and keep the power consumption constant. The V_{dac} will have considerable voltage variation which will create distortion.

Ideally the voltage drop is :

$$V_{DD} - V_{dac} = \alpha 2^N I_{LSB} R_{load}$$

$$I_{err} = \alpha 2^N V_{dac} / |Z_{cur}|$$

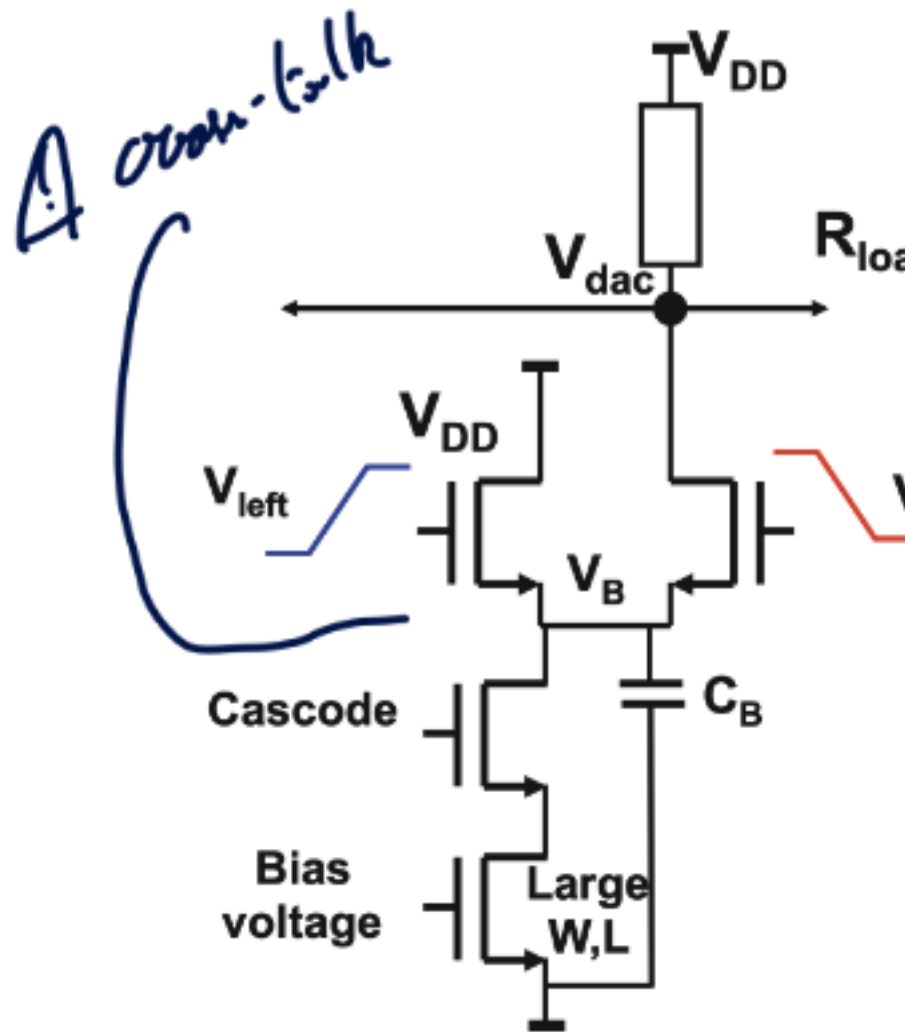
But we have a real voltage drop of :

$$V_{DD} - V_{dac} = R_{load} \left(\alpha 2^N I_{LSB} + \frac{\alpha 2^N V_{dac}}{|Z_{cur}|} \right)$$

$$V_{dac} = \frac{V_{DD} - R_{load} \alpha 2^N I_{LSB}}{1 + \alpha 2^N R_{load} / |Z_{cur}|}$$

$$\approx (V_{DD} - R_{load} \alpha 2^N I_{LSB}) \left(1 - \frac{\alpha 2^N R_{load}}{|Z_{cur}|} + \left(\frac{\alpha 2^N R_{load}}{|Z_{cur}|} \right)^2 \right)$$

We will have a HD3 with $((2^N R_{load}) / (4|Z_{cur}|))^2$



2.2.2.6 Switch implementation r0.3

Switches are controlled by low-swing differential pulses to reduce cross-talk and minimize the creation of an inversion charge. We need to minimize the V_B to have extremely accurate timing of switches. There is some charge taken from or released in output node at every switching event. Glitches are problematic for time-continuous output.

Architecture of choice for most demanding applications. We have a constant (high) power consumption and need a high-frequency clock and decoding logic require significant power. Differential implementation to cancel second-order distortion (big advantage). First-order relationship between linearity and bandwidth.

2.2.3 Charge domain

2.2.4 Time domain

2.3 Accuracy improvement methods

3 Nyquist ADC

This chapter will focus essentially about the comparator and will only cover some techniques applicable to Nyquist ADC.

3.1 Comparator

3.2 Comparator

3.2.1 Requirements and operation principle

A good starting point is to first summarize what we will need for this comparator block. Typically we want to measure and decide what an analog value should be in digital. So we will have to go from mV to V and be capable to be quite fast which means a good bw. We have to be quite accurate while not using too much power and be able to measure and transform a wide range of signal. We must also ensure to always have a clear decision (no metastability) and no memory effect. So quite challenging.

3.2.1.1 Limiting Amplifier We could think of chaining multiple amplifier but this can be quite slow and have some memory effect due to loss of inversion charge at output stages.

$$V_{out}(t) = A^M V_{in}(t=0) \left[1 - e^{-t/\tau} \sum_{i=1}^M \frac{(t/\tau)^{i-1}}{(i-1)!} \right] \approx A^M V_{in}(t=0) \sum_{i=1}^M \frac{(t/\tau)^i}{(i-1)!}$$

3.2.1.2 Latch behavior circuits We can also have some circuit that have a small feedback path which will create a hysteresis. It will first pre-amplify the small input signal and the positive feedback stage is activated to regenerate the still-small signal quickly.

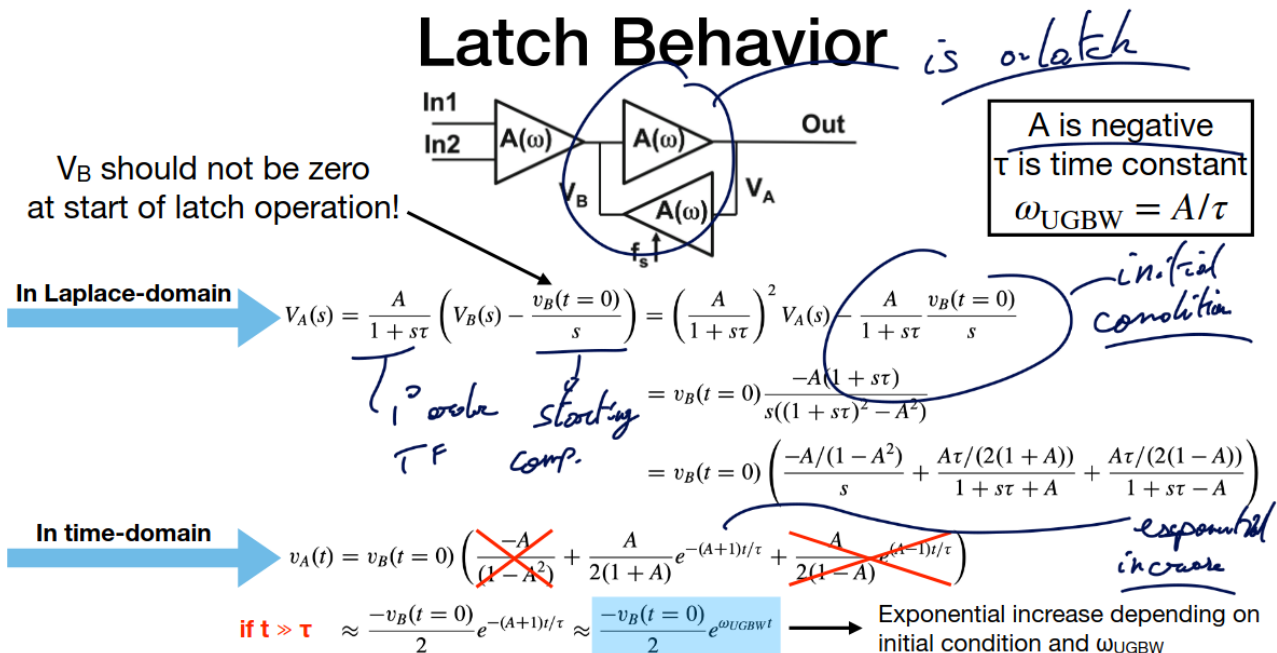


Figure 16: Latch behavior

3.2.2 Non-idealities

3.2.3 Comparator circuit examples

3.3 Analog-to-digital converter topologies

3.4 Conclusions

4 Delta Sigma adc

4.1 Basic principles

Table 4: Quick recap of definitions

Term	Definition
Input sampling rate	Frequency at which the analog input is sampled
Output data rate	Frequency at which updated output words are available
Conversion gain	Ratio of an output amplitude over the input amplitude
Bandwidth	Frequency span in which the amplitude of an output sine is less then 3dB lower as the input amp
snr	//
enob	Error free signal resolution = $(SNDR - 1.76 \text{ dB}) / (6.02 \text{ dB})$

All those values are important metric when specifying an adc but they can be misleading. We can trick the reader that we achieved exceptional result with a mediocre implementation. In fact We could find a hugeeee snr if we filter the noise at the input 5 times lower than the nyquist frequency and then claim a figure that is $\sqrt{5}$ better than what we would truly have.

4.1.1 fom

There is two main type of fom used for adc :

1. Schreier : It is based on a theoretical idea and on information theory. We have a signal power and a maximum efficiency. It is based on the *fundamental tradeoff* between accuracy and bandwidth, noise and power. But a critique that can be made is the fact it only uses snr and not the snr which take into account more realistic issue.
2. Walden : It takes into account the snr and also use the bandwidth instead of only using the Nyquist frequency. So a more band limited fom. The fom is in Joule per conversion step.

$$P_{sig,min} = 8kT \cdot BW \cdot SNR$$

$$FoM = 10^{10} \log \left(\frac{SNR \cdot BW}{P_{ADC}} \right)$$

$$\text{Power efficiency} = \frac{P_{circuit}}{P_{sig,min}} = \frac{P_{circuit}}{8kT \cdot BW \cdot SNR}$$

$$= SNR(dB) + 10^{10} \log \left(\frac{BW}{P_{ADC}} \right)$$

$$FoM = \frac{Power}{2^{ENOB} \cdot \min(2BW, f_s)}$$

There is also an issue related to the actual power consumption. In most paper, they are only taking into account the power at the input and not of the full system. In a SAR adc for example, V_{in} will just control some switches but not directly charge or discharge the cap. SAR ADC are not that power efficient after all if we take into account the external bench power supply that is required to let this ADC run.

4.1.1.1 Signal conditioning Again from information theory, we can derive a relation with a technological constant C_T .

$$\frac{speed(accuracy)^2}{power} = C_T$$

4.1.2 Resolution

The resolution is the *smallest discrete step* a system can take. Watch out ! resolution $\neq$ accuracy. We can have a granularity of 16 bits but being unable to provide 16 bits of information (2 bits tied to ground, ...).

$$resolution = \frac{\Delta I_R}{I_{max} - I_{min}} \cdot 100\%$$

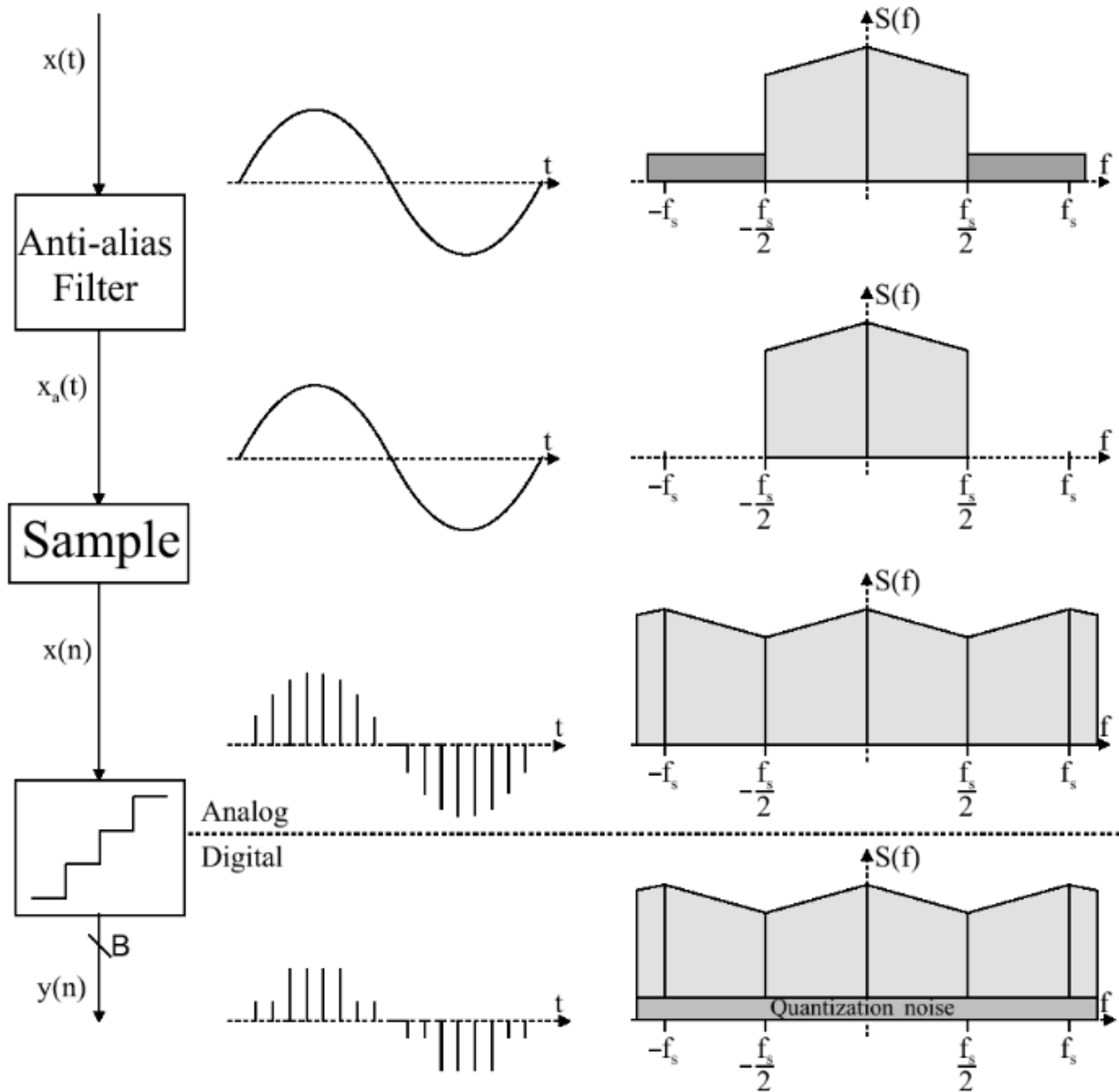


Figure 17: The conversion chain

We first remove the alias, then sample it to go into discrete time. We then quantize sampled data which will inevitably introduce quantization error. Finally to reconstruct we will use a *decimation filter*. In a Sigma-delta adc we will use something called the *osr* to improve this quantization noise.

$$OSR = \frac{\text{sampling freq } f_s}{\text{Nyquist freq } 2f_m}$$

For the n -bit quantizer, it will introduce this infamous quantizer error that have a seesaw pattern up until the overload area where it can no longer sample anything. This error is :

$$|V_{in}| < x_{max} \quad |Q_{error}| < \frac{\Delta}{2} \quad |V_{in}| > x_{max} \quad \text{overload}$$

The quantization noise is a purely *mathematical model*, it is not stochastic over time. It represents this transformation of a random input signal into a random output error due to quantization.

We can do the white noise approximation for fast varying signal where the error is evenly distributed between $[-\Delta/2; \Delta/2]$ with a pdf value of $1/\Delta$. We need no overload and need a change in input.

$$\begin{aligned} E(e_q) &= \frac{1}{\Delta I_R} \int_{-\frac{\Delta I_R}{2}}^{\frac{\Delta I_R}{2}} e de = 0 \quad \sigma_q^2 = \frac{1}{\Delta I_R} E((e_q - E(e_q))^2) \\ &= \frac{1}{\Delta I_R} \int_{-\frac{\Delta I_R}{2}}^{\frac{\Delta I_R}{2}} e^2 de \\ &= \frac{\Delta^2 I_R}{12} \end{aligned}$$

The power is the *variance* of the signal, which is equivalent to integrated noise. We only calculate the noise for a specific band.

$$\begin{aligned} x_{\max} &= 2^B \frac{\Delta}{2} \frac{1}{k} \\ S &= y_{\text{rms}}^2 = (k x_{\text{rms}})^2 = k^2 \frac{x_{\max}^2}{2} \\ &= 2^{2B-3} \Delta^2 \\ N &= \frac{\Delta^2}{12} \\ \Rightarrow SNR &= \frac{12}{8} 2^{2B} = \frac{3}{2} 2^{2B} = SNR_p = 1.76 + 6.02B \text{ dB} \end{aligned}$$

By going over the sampling much more often we will spread this noise leading to better SNR performance.

$$\begin{aligned} N_q &= \int_{-\frac{f_s}{2}}^{\frac{f_s}{2}} S_e^2(f) |H_e^2(f)| |H_d^2(f)| df = \int_{-f_b}^{f_b} h_e^2(f) df = \frac{\Delta^2}{12} \frac{2f_b}{f_s} \\ &= \frac{\Delta^2}{12 OSR} \end{aligned}$$

$$SNR_p = \frac{3}{2} 2^{2B} OSR \quad SNR_p = 1.76 + 6.02B + 10 \log(OSR) \text{ dB}$$

The main idea being delta sigma adc is to take advantage of previous error and to integrate it (PI) to correct and go to better values. The *delta* modulation comes from measuring the error and subtracting its output from the input. The storing (integrating in continuous time) is the principle of the *delta-sigma* modulation.

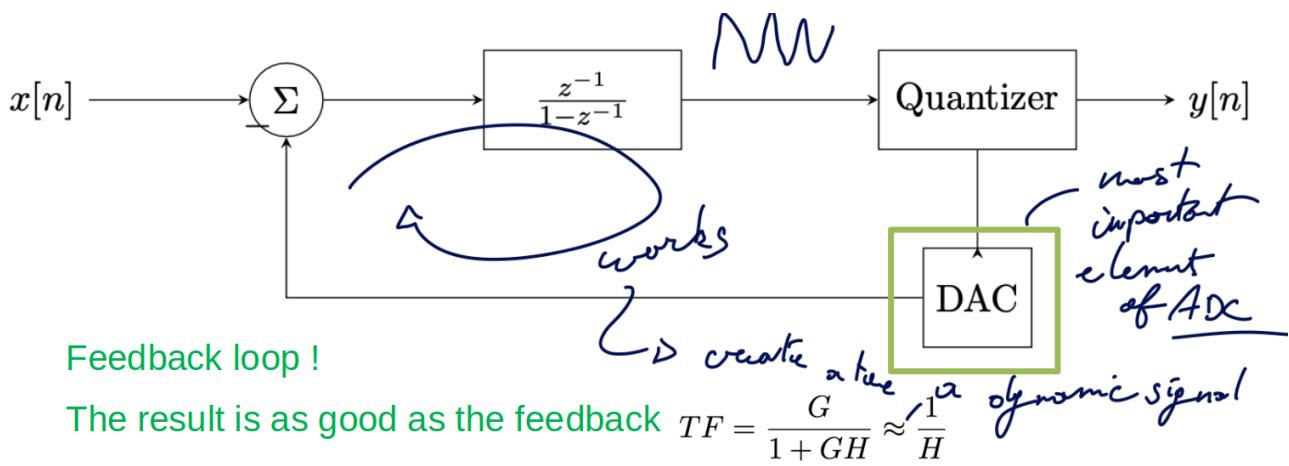


Figure 18: Delta sigma integration

For the quantizer we can simply go for a comparator so 1 or -1 and this is inherently linear. This will give us a noise and signal transfer function :

$$\begin{aligned}
 STF &= \frac{y}{x} = \frac{G}{1 + GH} \\
 &= \frac{k \frac{1}{z-1}}{1 + k \frac{1}{z-1}} \\
 &= \frac{kz^{-1}}{1 - z^{-1} + kz - 1} \\
 &= z^{-1} \quad \text{for } k = 1
 \end{aligned}
 \qquad
 \begin{aligned}
 NTF &= \frac{y}{e} = \frac{1}{1 + GH} \\
 &= \frac{1}{1 + k \frac{1}{z-1}} \\
 &= \frac{z - 1}{z + (k - 1)}
 \end{aligned}$$

We can clearly see the different transfer function which is a key aspect of delta sigma, it will do what we call *noise shaping*. To get the continuous transfer function we can replace z by $e^{j2\pi f/f_s}$. Again to find the SNR we have :

$$\begin{aligned}
 N_Q &= \int_{-f_b}^{f_b} \frac{\Delta^2}{12f_s} NTF^2 df = \frac{\Delta^2}{4} \left(\frac{\pi}{3}\right)^2 \left(\frac{1}{OSR}\right)^3 \\
 S &= \frac{\Delta 2^B}{2\sqrt{2}} \\
 SNR_{dB} &= 10 \cdot \log(S) - 10 \cdot \log(N_Q) = 6.02B - 3.41 + 30 \cdot \log(OSR)
 \end{aligned}$$

One major approximation that this formula relies on is the $\sin\left(\frac{2\pi}{OSR}\right) \approx \frac{2\pi}{OSR}$. So that's why we need high osr values or we won't take advantage of this noise shaping.

4.1.3 Second order Sigma Delta

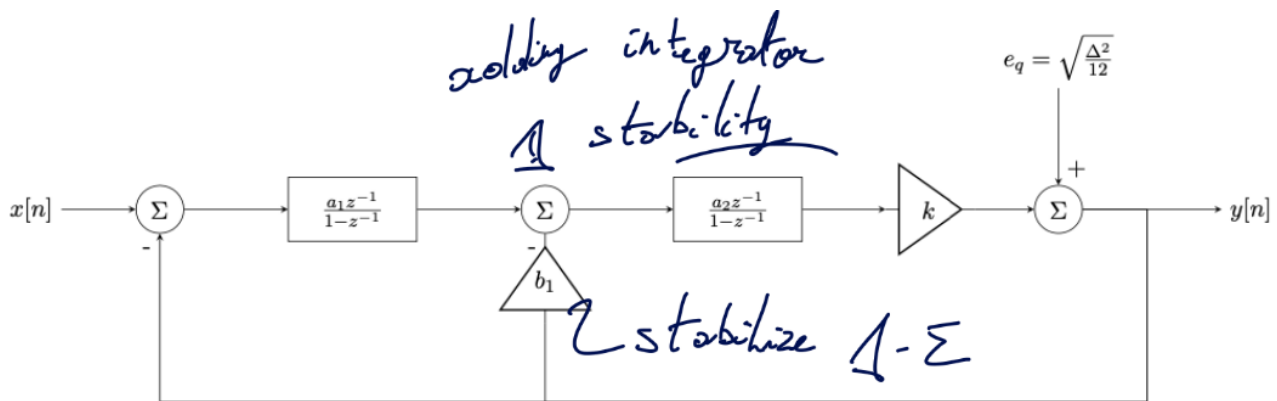
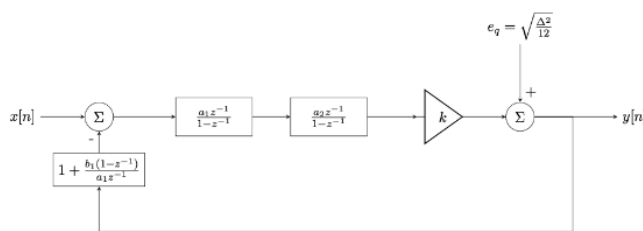


Figure 19: Second order delta sigma

We can indeed have better noise shaping but we need to take into account potential instability of this loop. We need to pick the right b_1 for the job.

Equivalent block diagram



$$NTF = \frac{1}{1 + GH}$$

gain of 1

$$\begin{aligned}
 &= \frac{1}{1 + k \frac{a_1 z^{-1}}{1-z^{-1}} \frac{a_2 z^{-1}}{1-z^{-1}} \frac{a_1 z^{-1} + b_1(1-z^{-1})}{a_1 z^{-1}}} \\
 &\quad \text{take } a_1 = 1, a_2 = 1, b_1 = 2, k = 1 \\
 &= \frac{1}{1 + \frac{z^{-1} + 2(1-z^{-1})}{1-z^{-1}} \frac{z^{-1}}{1-z^{-1}}} \\
 &= \frac{(1-z^{-1})^2}{(1-z^{-1})^2 + (z^{-1} + 2(1-z^{-1}))z^{-1}} \\
 &= \frac{(1-z^{-1})^2}{1 - 2z^{-1} + z^{-2} + z^{-2} + 2z^{-1} - 2z^{-2}} \\
 &= (1-z^{-1})^2
 \end{aligned}$$

Figure 20: Equivalent second order

Again, we can repeat a similar process as in equation [eq:snr-noise] and have :

$$\begin{aligned}
 SNR &= 6.02B - 11.1 + 50 \cdot \log(OSR) \\
 SNR_n &= 6.02B - 10 \cdot \log\left(\frac{2\pi^{2n}}{3(2n+1)}\right) + 10(2n+1) \cdot \log(OSR)
 \end{aligned}$$

We can also have other issues that will limit our enob, typically we can have third order distortion. But again, some paper will cheat and move the f_{in} as close as the cut-off frequency so the third order distortion won't be seen.

The main action of a delta sigma, is converting a signal into a pwm signal. But we could also face some saturation where the width of the signal doesn't reflect or mean any information. We need the amplitude to

be lower as the feedback voltage to avoid *overload*. Different order sigma delta have some different overload criteria, we will try to spread this swing to avoid overloading early stage integrator.

4.1.4 Stability and Limit Cycle

We know it is a feedback system so having the wrong coefficients will make an instable system. We can witness some *limit cycle* as starting with some starting condition will evolve with the same trajectory. The instability can be seen as a loop that will go towards infinity and won't find stability. Usually, this will be limited by the supply range.

4.1.4.1 Pattern noise There is a pattern in the ideal output at $f_s/4$ and is called the limit cycle. Cause of this, the quantization is no longer white and we cannot longer do noise shaping. Dead zone in DC transfer function. We can use some matlab models to check for stability.

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$$\nabla \times \left(\frac{1}{\mu} (\nabla \times \mathbf{A}) \right) = \mathbf{J} + \epsilon \frac{\partial}{\partial t} \left(-\nabla \Phi - \frac{\partial \mathbf{A}}{\partial t} \right)$$

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